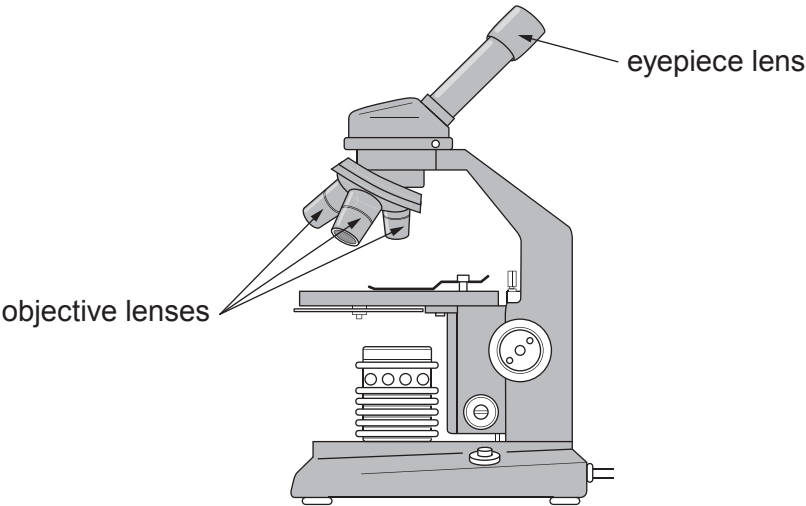


2. Rheinallt observed some plant cells using a light microscope.



(a) He set up the microscope so that it had a total magnification of $\times 240$.

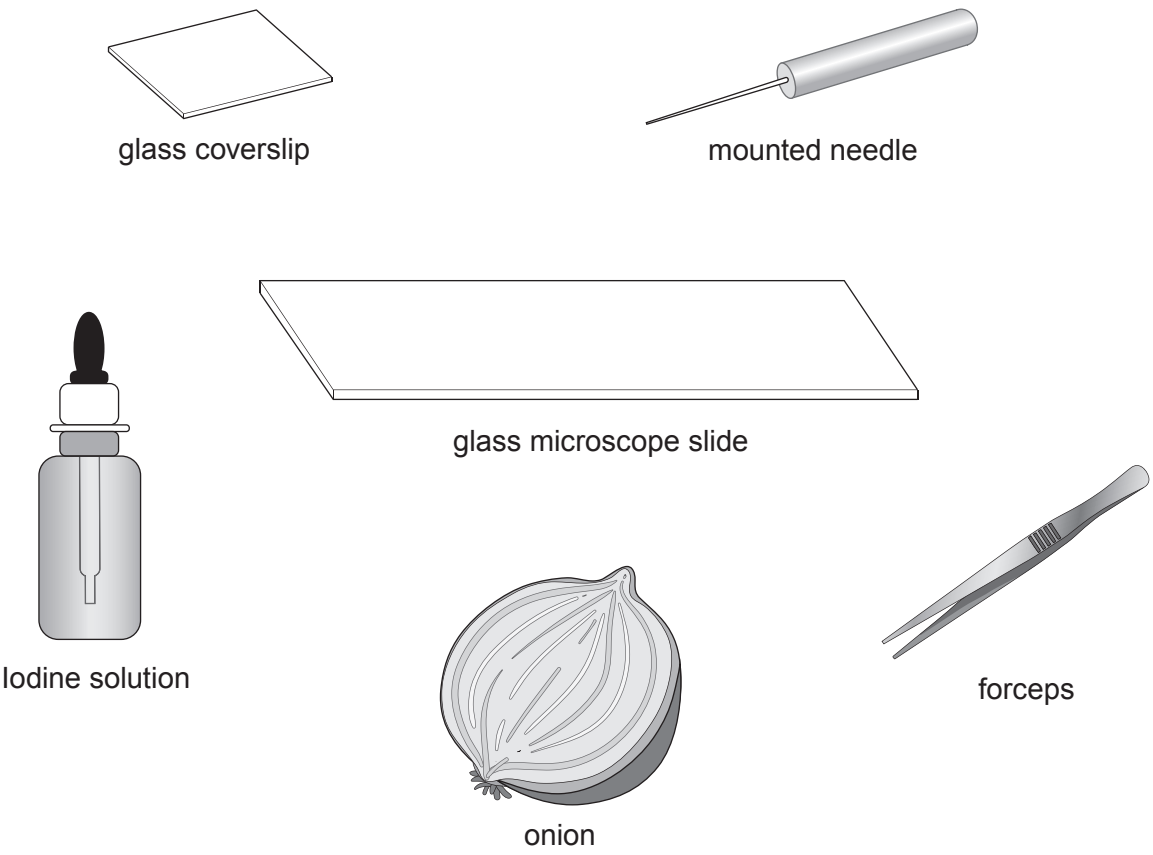
The magnification of the eyepiece lens on his microscope was $\times 6$. Which of the following objective lenses did he use? [1]

Objective lens power	Magnification
low	$\times 8$
medium	$\times 15$
high	$\times 40$

Objective lens magnification = $\times$



(b) The diagram shows the apparatus Rheinallt used to prepare a slide of onion cells.



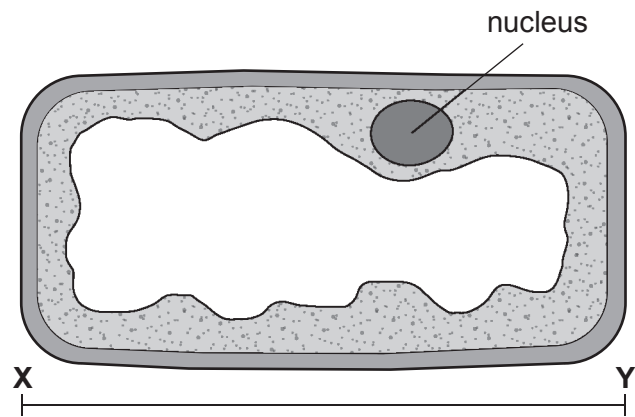
Continue and complete the method below, which he used to prepare his slide for observation under the microscope. [3]

Method

- Take a piece of freshly cut onion.
- Use forceps to carefully peel a thin layer of cells from the inner surface of the onion.
- Place the layer of onion cells onto the centre of a microscope slide.
-
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-



(c) Rheinallt made a large drawing of one of the cells he observed.



- (i) I. Measure the length of the cell in Rheinallt's drawing along line **X–Y**. [1]

Length of cell in drawing = mm

- II. The actual length of cells of this type is usually 0.02 mm.
Use your answer to part I. to calculate the magnification of Rheinallt's drawing. [2]

Magnification of drawing = $\times$

- (ii) From the drawing, give **one** feature of the cell which shows that it is a plant cell. [1]

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- (d) Rheinallt compared his drawing with an image obtained from an electron microscope.
Why would the electron microscope image give more information about the structure of an onion cell? [1]

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